

Segregated by Design in Brazil? Street-Network Topology and the Measurement of Racial Segregation

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Abstract

Spatial segregation indices usually define each resident’s local environment with straight-line distance, although people move along streets. Following Knaap and Rey (2024), we compare racial segregation measured with Euclidean and with pedestrian network distance for the 91 Brazilian municipalities with at least 300,000 inhabitants, using 2022 Census colour-or-race counts for 151,992 urban tracts and OpenStreetMap walking networks. The two local environments share an identical 2-km kernel and differ only in the distance metric. With network distance, the multigroup spatial information theory index is higher in every city, by 29.8% on average, and the ranking of the most segregated cities changes, with hilly and island cities such as Florianópolis and Vitória climbing. However, 2-km walking environments contain only about 50% of the population of 2-km circles. When Euclidean environments are shrunk to hold the same population, the gap vanishes (mean difference -1.6%). Across cities the gap and the share of reachable population are associated with dead-end streets, circuitry and meshedness, but not with a metric analogue of the space-syntax fragmentation that characterises Brazilian street grids. Street networks thus shape measured segregation mainly by setting the effective scale of neighbourhoods, which argues for population-matched comparisons and points to local street permeability as a planning lever. All analyses use the open-source PySAL **segregation** package and are fully reproducible.

Keywords: residential segregation; street networks; spatial weights; urban morphology; OpenStreetMap; Brazil; PySAL

1 Introduction

Straight-line distance is the default abstraction of applied spatial analysis. It is a useful simplification, but people do not move across a featureless plane: they walk along streets, around blocks, over bridges and up stairways, and the configuration of that network conditions whom they can easily meet. In segregation research the abstraction enters through the definition of the “local environment” that spatial indices compare with the city as a whole (Reardon and O’Sullivan, 2004). Most applications define that environment with Euclidean distance. Knaap and Rey (2024) showed, for every metropolitan area in the United States, that replacing Euclidean distance with walking distance along the street network raises the spatial information theory index in almost every metro, by about a fifth on average, and reshuffles the ranking of the most segregated places. They also showed that the size of the gap is associated with the topology of the street network.

This paper asks whether the same holds in Brazil, where there are strong reasons to expect the street network to matter even more. Brazilian cities combine colonial cores, modernist plans, sprawling peripheral *loteamentos*, self-built favelas on steep hillsides and, since the 1990s, gated condominiums (*condomínios fechados*) that interrupt the public street grid (Coy, 2006). In the

most comprehensive configurational comparison to date, Medeiros (2013) found that the street networks of the 27 Brazilian state capitals are among the most fragmented of 164 cities worldwide: a “patchwork” of locally coherent pieces that are poorly articulated with one another. If segregation is, in part, a restriction on the possibility of interaction (Netto et al., 2015), then a patchwork city should look more segregated when the local environment is measured along the network than when it is measured as the crow flies.

We compute racial segregation for the 91 Brazilian municipalities with at least 300,000 inhabitants, using the 2022 Census colour-or-race counts for urban census tracts and pedestrian networks from OpenStreetMap. For each city we build two local environments that differ *only* in the distance metric: the same 2-km triangular kernel is applied to Euclidean and to network distances. We ask two questions. First, how different are segregation levels and rankings under the two metrics (RQ1)? Second, which properties of the street network are associated with the gap (RQ2)? To RQ2 we add a Brazil-specific predictor, a metric analogue of the space-syntax *synergy* measure that underlies Medeiros’s diagnosis of fragmentation. Because network environments contain fewer people than Euclidean environments of the same radius, part of any gap could be a scale effect rather than a barrier effect (Saporito, 2026); we therefore also report a population-matched comparison in which each tract’s Euclidean environment is shrunk until it holds as many people as its network environment.

The paper makes three contributions: the first nationwide Euclidean-versus-network comparison of segregation for a country of the Global South; a test of whether configurational fragmentation, a long-standing theme of Brazilian urban morphology, helps explain the gap; and a fully reproducible workflow built on the open-source PySAL `segregation` package (Cortes et al., 2026, 2020; Rey et al., 2021, 2022). Consistent with Knaap and Rey (2024), we treat the comparison descriptively; a formal inferential test of the gap is left for future work.

2 Background

2.1 Measuring segregation in space

Spatial segregation indices replace the question “who lives in the same tract?” with “who lives nearby?”. White (1983) introduced distance decay into the measurement of segregation, and Wong (1993) showed that indices based on the adjacency of areal units depend on unit size and shape. Reardon and O’Sullivan (2004) provided a general framework in which any theoretically motivated proximity function defines each person’s local environment; among the resulting measures, the spatial information theory index H has the most desirable properties. Reardon et al. (2008) used the bandwidth of the proximity function to describe the geographic scale of segregation. In Brazil, Feitosa et al. (2007) developed kernel-based global and local spatial indices and applied them to São José dos Campos. In all of these, proximity is almost always Euclidean.

2.2 Street networks and social interaction

A second literature argues that street networks structure the opportunities for contact. Grannis (1998) showed that residential (“trivial”) streets bound neighbourhood networks, and Grannis (2005) that pedestrian-connected “T-communities” predict racial composition in three US cities. Roberto (2018) incorporated road connectivity into segregation measurement and found higher segregation in Pittsburgh under network distance. Knaap and Rey (2024) scaled this comparison to all US metros and related the gap to network topology measured with OSMnx (Boeing, 2017) and momepy (Fleischmann, 2019): circuitry widens the gap, intersection density narrows it, and

cyclomatic complexity (redundancy) dampens the effect of circuitry. Global comparisons show that street-network form varies far more outside North America: grids are the exception, and circuitry and orientation entropy are higher (Boeing, 2019, 2022). Recently, Saporito (2026) cautioned that network-based environments of a given radius contain fewer people, so that part of the network–Euclidean difference may reflect the population size of local environments rather than barriers to movement.

2.3 The configuration of Brazilian street networks

Brazilian research on urban form has long emphasised configuration. Drawing on space syntax, Holanda (2002) interpreted the plan of Brasília as a “space of exception” whose dispersed, hierarchical layout separates social groups. Medeiros (2013) compared axial maps of 164 cities, including all 27 Brazilian capitals, and concluded that Brazilian cities are exceptionally fragmented: patchworks of internally coherent pieces whose local and global structures are poorly articulated (low *synergy*), a labyrinthine configuration that hampers movement between parts of the city. Netto et al. (2015) reframed segregation as a restriction on interaction produced jointly by residential location, the street network and mobility, with evidence from a Brazilian city. Two recent studies bring the street grid directly into segregation measurement: Saboya and Peres (2025) propose CHASM, an exposure measure computed on street segments, and Gonçalves et al. (2024) use a network-based potential-accessibility index to reveal racial and income segregation patterns in Pelotas. At the scale of informal settlements, Carvalho and Netto (2023) show that favelas are themselves internally segregated. Using OSM, Spadon et al. (2018) characterised the street networks of the 645 municipalities of São Paulo state by topological features. Finally, the spread of gated condominiums has privatised parts of the street network and fragmented the urban fabric (Coy, 2006). Together these works motivate three expectations: (H1) segregation is higher under network than under Euclidean distance; (H2) the gap is larger where circuitry, dead-end streets and configurational fragmentation are higher; and (H3) the gap is smaller where intersection density and meshedness are higher.

2.4 Race, space and access in Brazilian cities

Racial residential segregation in Brazil is moderate by US standards but real and entangled with class (Telles, 2004); a nationwide analysis of 2010 census tracts found it highest in the South and Southeast (Sousa Filho et al., 2023). The consequences of location are mediated by the transport network. The IPEA *Acesso a Oportunidades* project shows that white and high-income residents of the 20 largest Brazilian cities reach more jobs, schools and health services than black and low-income residents by every transport mode (Pereira et al., 2019b; Tomasiello et al., 2024), and that transport policy can widen these gaps (Pereira et al., 2019a). Bittencourt et al. (2021) show that accessibility and segregation reinforce each other in four metropolises, and Boisjoly et al. (2020) document large accessibility differences across four cities. This literature measures travel along networks (Pereira et al., 2021) but takes residential segregation as given; our study brings the network into the measurement of segregation itself.

3 Data and methods

3.1 Cities, tracts and population groups

The study covers the 91 municipalities with a 2022 Census population of at least 300,000, a set that includes all 27 state capitals. The spatial units are 2022 census tracts (*setores censitários*). We keep tracts classified as urban (*SITUACAO = Urbana*) with at least one resident, 151,992 tracts holding 98.9% of the population of the 91 municipalities. We restrict the analysis to urban tracts because several municipalities (e.g. Porto Velho or Manaus) contain very large rural tracts whose centroids lie kilometres from any street, which would inflate network distances for reasons unrelated to urban design. Population counts come from the Census tract aggregates for colour or race: *branca*, *preta*, *amarela*, *parda* and *indígena*. Cells suppressed for confidentiality are set to zero.

3.2 Street networks

For each municipality we download the OpenStreetMap pedestrian network (all ways usable on foot, including footpaths and stairways) with OSMnx (Boeing, 2017) for the union of its urban tracts buffered by 2 km, so that walking paths may leave the urban footprint. Graphs are simplified, projected to the SIRGAS 2000 UTM zone of the city, made undirected, and reduced to their largest connected component. Each tract is represented by a point guaranteed to lie inside the polygon and snapped to the nearest network node; the snap distance is kept and added to network distances.

3.3 Local environments and the segregation index

Let d_{ij} be the distance between tracts i and j . We use two distances: Euclidean d_{ij}^E , and network $d_{ij}^N = s_i + \ell(n_i, n_j) + s_j$, where n_i is the node nearest to tract i , s_i the snap distance and ℓ the shortest-path length. Because the snapped path is itself a path between the two points, $d_{ij}^N \geq d_{ij}^E$. Both distances enter the same triangular kernel,

$$w_{ij} = \begin{cases} 1 - d_{ij}/b & \text{if } d_{ij} \leq b, \\ 0 & \text{otherwise,} \end{cases} \quad w_{ii} = 1, \quad (1)$$

with bandwidth $b = 2,000$ m, as in Knaap and Rey (2024). The local environment of tract i is the spatially weighted population $\tilde{\tau}_{im} = \sum_j w_{ij} \tau_{jm}$ of each group m , and $\tilde{\pi}_{im} = \tilde{\tau}_{im} / \sum_m \tilde{\tau}_{im}$. The entropy of each local environment is $\tilde{E}_i = -\sum_m \tilde{\pi}_{im} \ln \tilde{\pi}_{im}$, and the spatial information theory index is

$$H = 1 - \frac{1}{TE} \sum_i \tilde{t}_i \tilde{E}_i, \quad (2)$$

where $\tilde{t}_i$ is the local population, T the total population and E the city-wide entropy (Reardon and O’Sullivan, 2004). H equals 0 when every local environment has the city’s composition and 1 when every local environment contains a single group. We compute H over the five colour-or-race groups (a group absent from a city contributes zero). As secondary measures we compute the spatial versions of dissimilarity, isolation and entropy for the *preta+parda* population. All indices are computed with the PySAL `segregation` package (Cortes et al., 2026, 2020). We build the weighted local environments with a sparse matrix product identical to the one the package applies to a `libpysal` weights object, which a unit test verifies to machine precision, and then call the package’s estimators. For each city we record H^E , H^N , the gap $\Delta H = H^N - H^E$ and the relative gap $\% \Delta H = 100 \Delta H / H^E$.

Population-matched environments. Because $d_{ij}^N \geq d_{ij}^E$, a network environment of radius b always contains fewer (weighted) people than a Euclidean one, and smaller environments are, other things equal, less diverse. Following the concern raised by Saporito (2026), we also compare H^N with a Euclidean index computed from *population-matched* environments: for each tract we find, by bisection, the Euclidean radius $b_i \leq 2,000$ m at which the kernel-weighted population of its Euclidean environment equals that of its 2-km network environment, and recompute H with these tract-specific radii. Any remaining gap cannot be attributed to the number of people in local environments.

3.4 Network topology and configurational fragmentation

Topology is measured on all network nodes lying inside the urban footprint (the union of populated urban tracts). We keep every piece rather than only the largest component, because planned cities such as Palmas consist of superblocks that connect through streets outside the footprint. Table 1 lists the measures, which follow Knaap and Rey (2024). Densities are expressed per km² of footprint.

Table 1: Street-network topology measures (v nodes, e edges, p connected components).

Measure	Definition
Intersection density	nodes with degree ≥ 3 per km ²
Street density	km of street per km ²
Mean street length	mean edge length (m)
Streets per node	mean node degree (parallel edges counted)
Circuitry	$\sum$ edge length / $\sum$ straight-line length between edge end-points
Dead-end, 3-way, 4-way shares	share of nodes with degree 1, 3 and 4
Cyclomatic number	$e - v + p$ (independent loops; network redundancy)
Meshedness	$(e - v + p) / (2v - 5)$ (share of the maximum possible number of loops)
Gamma	$e / 3(v - 2)$
Fragmentation F	$1 - \text{corr}(\ln L_i, G_i)$, see text

Configurational fragmentation. Space syntax diagnoses a “patchwork” city by the weak correlation (low *synergy*) between the local and global integration of its street segments (Medeiros, 2013). Axial maps are not available for 91 cities, so we compute a metric analogue on the OSM graph. For a node i , local reach L_i is the total street length whose two endpoints lie within 800 m of network distance from i . Global closeness G_i is the reciprocal of the mean network distance from i to 500 randomly drawn urban nodes. Synergy is the Pearson correlation between $\ln L_i$ and G_i over 2,000 randomly drawn urban nodes, and fragmentation is $F = 1 - \text{synergy}$. Routing uses the full buffered network. High F means that locally well-connected places are not necessarily well connected to the city as a whole. We stress that F approximates the space-syntax measure with metric distances rather than replicating it.

3.5 Analytical strategy

For RQ1 we describe the distributions of H^E , H^N , ΔH and $\% \Delta H$, their Pearson and Spearman correlations and the change in the ranking of the most segregated cities, and we repeat the com-

parison with bandwidths of 1 and 3 km, an exponential kernel ($w = e^{-3d/b}$), the secondary indices and the population-matched environments. For RQ2 we estimate, across cities,

$$y_c = \alpha + \mathbf{x}_c' \boldsymbol{\beta} + \varepsilon_c, \quad (3)$$

with $y_c \in \{\Delta H, \% \Delta H\}$. $\mathbf{x}_c$ contains log intersection density, circuitry, meshedness, log cyclomatic number, dead-end share, log population density and H^E , plus the cyclomatic $\times$ circuitry interaction of Knaap and Rey (2024). We then add F and compare the fit. Normally distributed regressors are standardised and skewed ones logged. Standard errors are heteroskedasticity-robust (HC3). With $N = 91$ the models are descriptive associations, not causal estimates.

4 Results

4.1 How much does the distance metric matter?

Figure 1 illustrates the difference between the two local environments for three contrasting fabrics. In Goiânia’s planned grid the area reachable within 2 km on foot fills much of the 2-km circle. In the hillside neighbourhoods of Rio de Janeiro and in Salvador, whose street network is organised around ridges and valleys, it covers a small, irregular part of it.

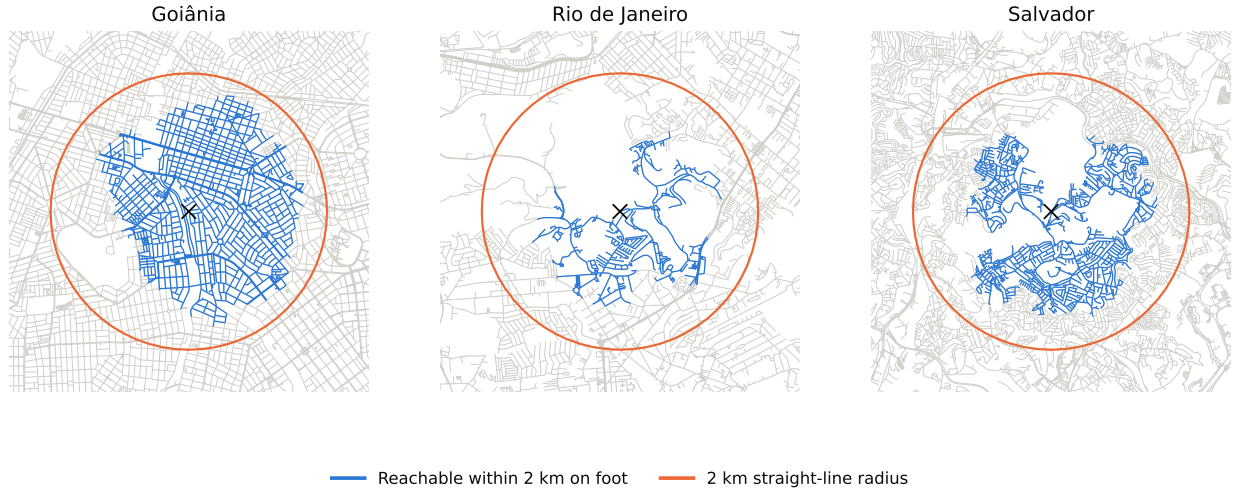


Figure 1: Street segments reachable within 2 km on foot (blue) and the 2-km straight-line radius (orange) from the node nearest the population-weighted centre of each city’s urban tracts.

Table 2 summarises the comparison for the main specification. Measured with network distance, the spatial information theory index is higher in all 91 municipalities (100.0%). The mean index rises from 0.022 to 0.028, a mean relative increase of 29.8%, larger than the $\approx 20\%$ reported for US metropolitan areas by Knaap and Rey (2024). The relative gap ranges from 0.8% to 109.0%. Levels of racial segregation measured by H are low in absolute terms, as expected for a five-group index in cities where most residents are *pardo* or white. The two measures are strongly correlated across cities (Pearson 0.980, Spearman 0.984, Figure 2).

Despite the high correlation, the choice of metric changes the answer to the question most often asked of segregation indices: which places are most segregated. Among the cities that rank in the top 15 under either metric, 11 change position (Figure 3). The largest moves are cities built on hills, islands and narrow valleys: Florianópolis rises from 38th to 13th, Juiz de Fora from 22nd to

Table 2: Spatial information theory index under Euclidean and network distance (2-km linear kernel, 91 municipalities). $\text{Gap} = H^N - H^E$; $\text{Gap (\%)} = 100 (H^N - H^E)/H^E$.

	count	mean	std	min	25%	50%	75%	max
H, Euclidean	91.000	0.022	0.017	0.001	0.009	0.017	0.032	0.072
H, network	91.000	0.028	0.020	0.002	0.011	0.022	0.040	0.079
Gap	91.000	0.005	0.005	0.000	0.002	0.004	0.007	0.026
Gap (%)	91.000	29.835	18.968	0.751	16.294	25.209	39.522	108.987

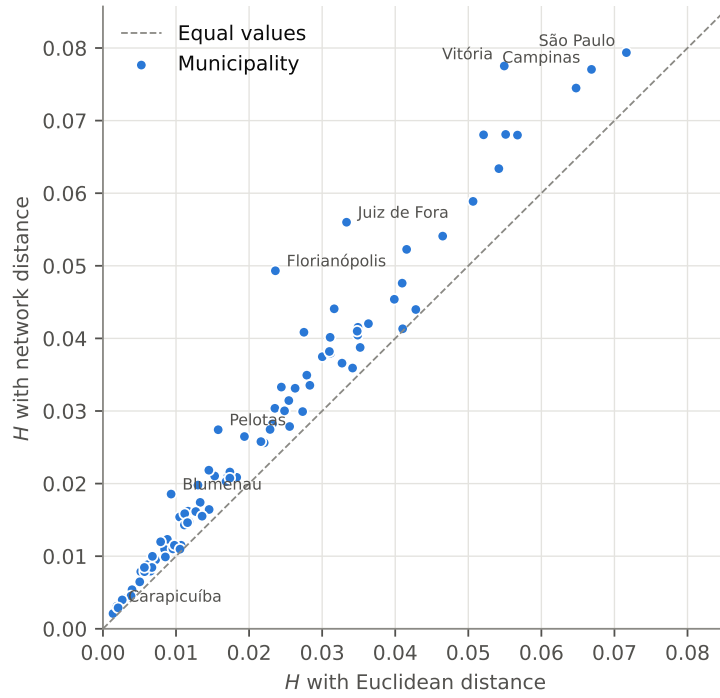


Figure 2: H under Euclidean and network distance. Points above the dashed line are cities where network distance yields higher segregation. Labelled: the five largest relative gaps and the three highest network values.

10th and Vitória from 6th to 2nd. Brasília (11th to 17th) and Santos (13th to 20th) fall. The map in Figure 4 shows no simple regional pattern in the relative gap. Large gaps occur in the South (Florianópolis, Blumenau, Pelotas), in Minas Gerais (Juiz de Fora), in peripheral municipalities of the São Paulo and Belo Horizonte metropolitan areas (Carapicuíba, Diadema, Ribeirão das Neves) and in Maceió.

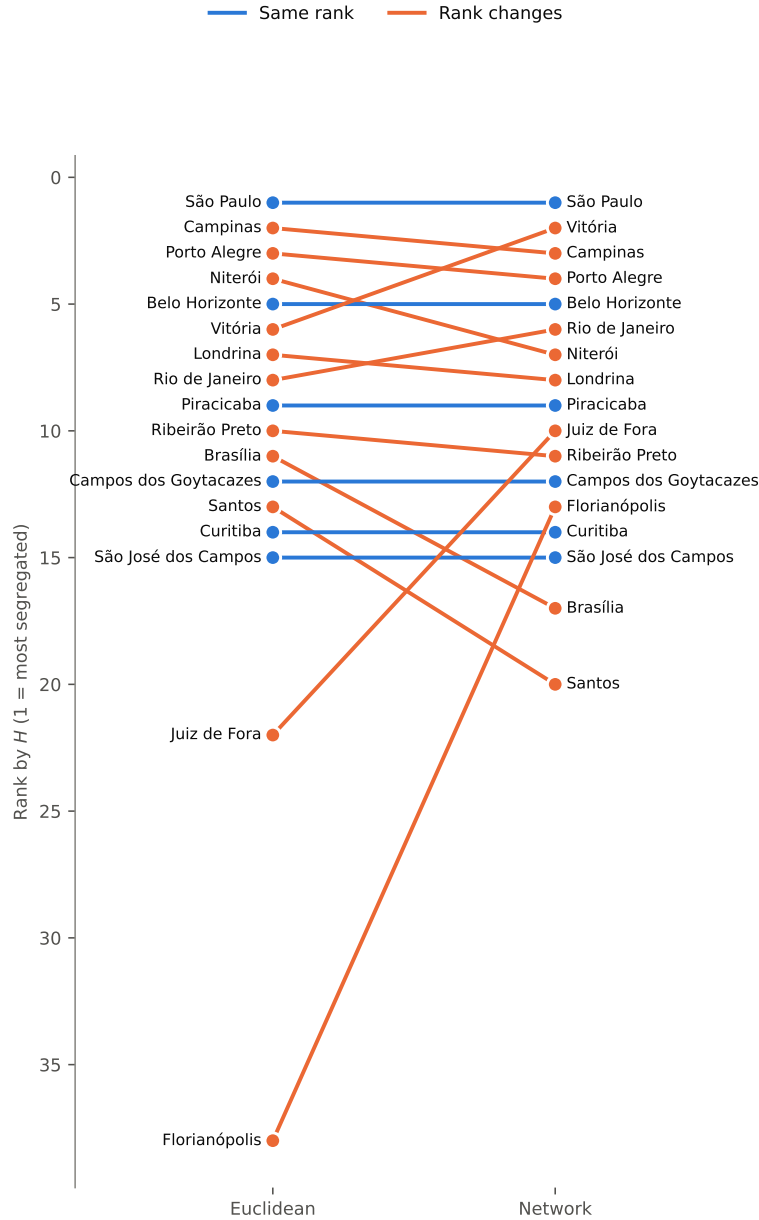


Figure 3: Ranks of the most segregated municipalities (top 15 under either metric) with Euclidean and network distance. Orange lines mark cities whose rank changes.

These results are robust to the kernel. With 1-km or 3-km bandwidths, an exponential kernel, or the single-group dissimilarity and entropy indices for the *preta+parda* population, network-based segregation remains higher in almost every city (Appendix Table 4). The exception is the isolation index, which changes little under either metric (its mean relative difference is below 2% in every

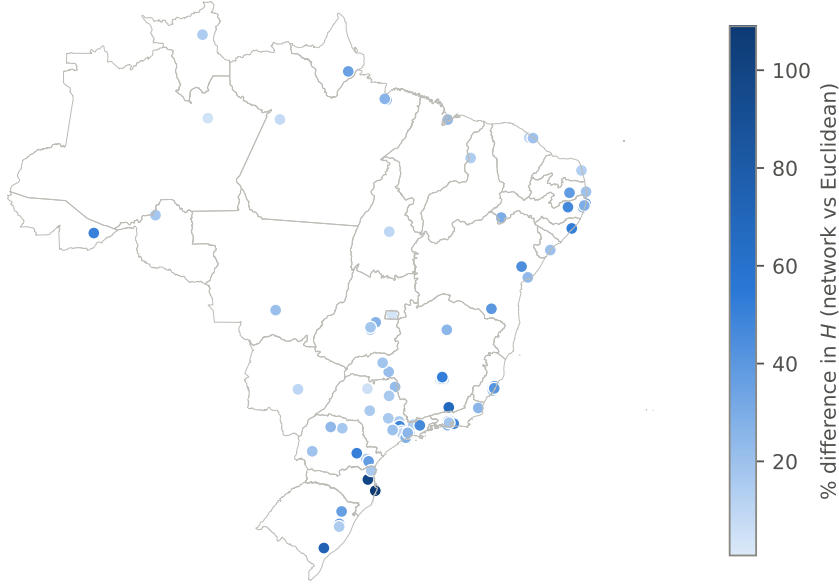


Figure 4: Relative difference between network and Euclidean H (%) by municipality.

specification).

Population-matched environments. They are not robust, however, to holding the size of local environments constant. Walking reaches far fewer people than a straight line: the mean 2-km network environment contains 50% of the (kernel-weighted) population of the corresponding Euclidean environment (median across cities), with a range of 32–59%. When each tract’s Euclidean radius is shrunk until its environment holds the same population as its network environment (median radius 1,330 m), the gap disappears. The mean relative difference becomes -1.6%, network H exceeds the matched Euclidean H in only 20.9% of cities, and the ranking under the two metrics becomes practically identical (Spearman 0.999). The same holds for the single-group indices (Table 4). In Brazilian cities, therefore, the network–Euclidean gap is almost entirely a difference in the *scale* of local environments, as Saporito (2026) argued for US cities, rather than evidence that the street network separates groups beyond what a smaller neighbourhood would.

4.2 Which network characteristics explain the gap?

Figure 5 shows the correlation structure of the topology measures. As in Knaap and Rey (2024), one cluster groups meshedness, gamma, streets per node and the share of 4-way intersections (connectivity). A second groups circuitry, dead-end share and self-loops (a dendritic, winding fabric). A third groups size and density. Configurational fragmentation F is nearly uncorrelated with all of them. The relative gap belongs with the dendritic cluster: it rises with the dead-end share (Spearman 0.40) and is essentially unrelated to F (-0.12).

Table 3 reports the regressions. In all four models the dead-end share is positively associated with both the absolute and the relative gap, and the level of Euclidean segregation is strongly associated with the absolute gap (positively) and the relative gap (negatively). The negative association with the relative gap means that low-segregation cities show larger relative differences,

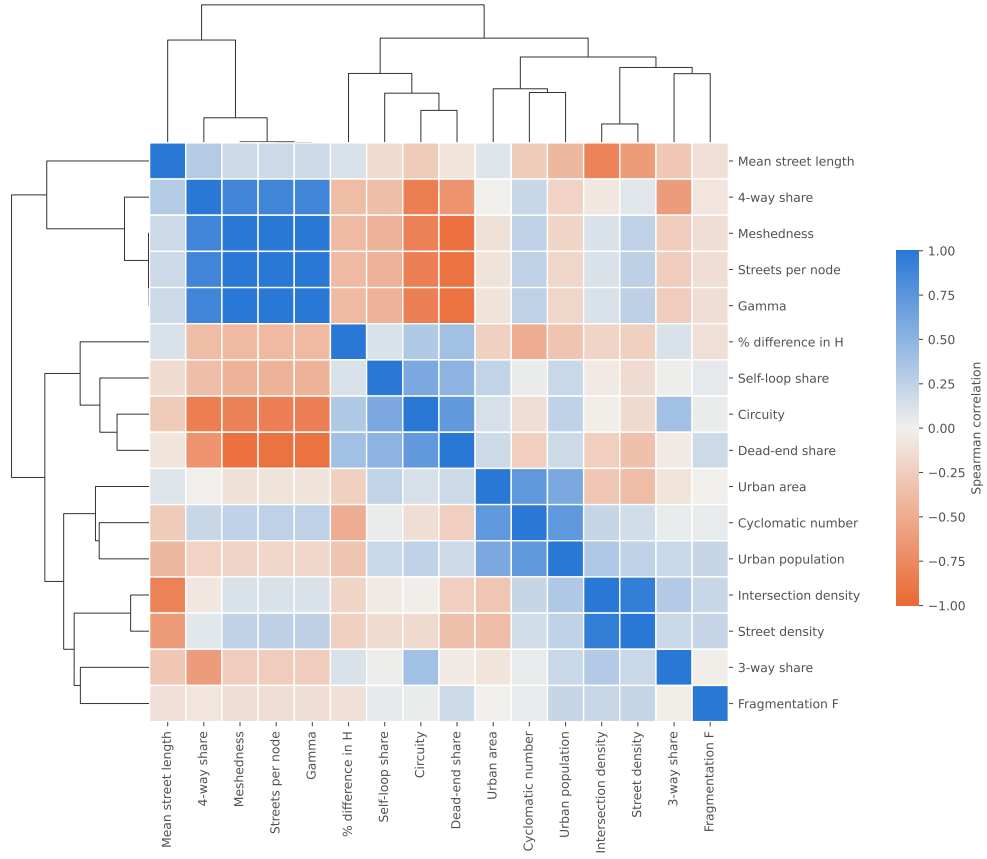


Figure 5: Spearman correlations among street-network topology measures, city size and the relative gap in H , hierarchically clustered.

although the absolute differences there are small. The cyclomatic number is negatively associated with the relative gap, consistent with more redundant networks approximating unconstrained movement. Circuity, meshedness, intersection density and the cyclomatic $\times$ circuity interaction are not significant once the dead-end share is included. Meshedness and the dead-end share are strongly collinear (variance inflation factors of 12.7 and 9.4), so their separate coefficients should be read with caution. Adding fragmentation F does not improve the fit: its coefficient is small, negative and not significant, and the adjusted R^2 falls slightly. Figure 6 makes the same point visually.

Table 3: OLS models of the absolute (ΔH) and relative ($\%\Delta H$) gap. K&R: the specification of Knaap and Rey (2024); $+F$: adding configurational fragmentation. Standardised (z) or logged regressors; HC3 standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	ΔH (K&R)	ΔH (+ F)	$\%\Delta H$ (K&R)	$\%\Delta H$ (+ F)
Intersection density (log)	0.0031 (0.0038)	0.0030 (0.0039)	14.34 (9.46)	13.66 (9.39)
Circuity	0.0001 (0.0072)	-0.0004 (0.0069)	21.36 (25.71)	15.51 (24.12)
Meshedness	0.0015 (0.0018)	0.0016 (0.0018)	-0.21 (5.59)	-0.10 (5.45)
Cyclomatic number (log)	-0.0017 (0.0011)	-0.0017 (0.0011)	-6.75** (3.36)	-6.38* (3.34)
Dead-end share	0.0025** (0.0012)	0.0026** (0.0012)	9.14** (4.46)	9.68** (4.30)
Population density (log)	-0.0027 (0.0022)	-0.0025 (0.0023)	-11.69* (6.79)	-10.15 (6.80)
H , Euclidean	0.0030*** (0.0005)	0.0030*** (0.0006)	-6.29*** (1.69)	-6.40*** (1.66)
Cyclomatic (log) $\times$ circuity	0.0001 (0.0008)	0.0002 (0.0007)	-2.23 (2.87)	-1.58 (2.68)
Fragmentation F		-0.0002 (0.0004)		-2.16 (1.65)
Intercept	0.0289** (0.0115)	0.0277** (0.0127)	121.95** (47.90)	109.01** (50.25)
N	91	91	91	91
Adj. R^2	0.481	0.476	0.466	0.471

The population-matched results suggest how topology matters. The share of the Euclidean population that is reachable on foot is itself strongly associated with network form: it is lower where dead ends (-0.65) and circuitous streets (-0.63) are common and higher where the network is meshed (0.69), but it is unrelated to fragmentation (0.09). The gap is in turn larger where that share is smaller (-0.40). Street topology thus shapes measured segregation mainly by setting the effective scale of the neighbourhood.

5 Discussion

Our first result replicates, and amplifies, the central finding of Knaap and Rey (2024). When local environments are defined by walking distance, racial segregation in large Brazilian cities is higher in every case, by 29.8% on average, and the ranking of the most segregated places changes. Cities built on hills, islands and valleys (Florianópolis, Vitória, Juiz de Fora) climb the ranking. Analysts who compare cities with Euclidean spatial indices would therefore reach different conclusions about

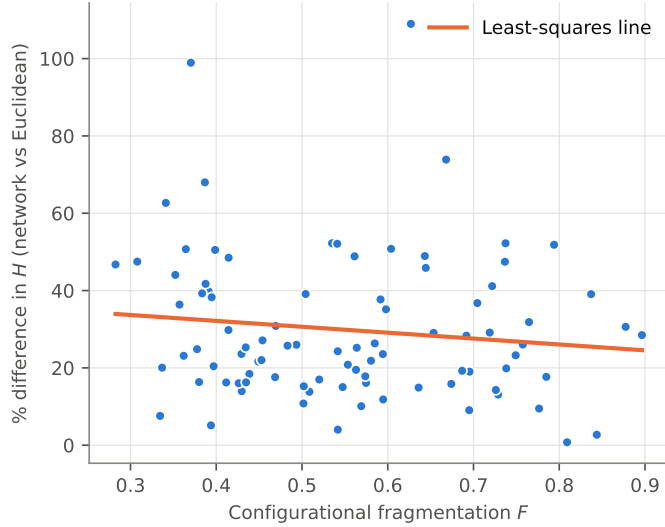


Figure 6: Configurational fragmentation F and the relative gap in H .

where segregation is most intense. This is consistent with the Brazilian configurational literature, which describes Brazilian street networks as unusually fragmented and labyrinthine (Medeiros, 2013; Holanda, 2002).

Our second result qualifies the first. Once the population of local environments is held constant, the gap vanishes. Walking along the network does not reveal a different social composition next door. It reveals a smaller neighbourhood, and smaller neighbourhoods are, as the multiscale literature shows, more segregated (Reardon et al., 2008). This supports the concern raised by Saporito (2026) and changes how the network–Euclidean gap should be read. It is not evidence that streets act as barriers that sort groups within a given population radius. It is evidence that street networks set the *effective scale* at which residents experience their city: in dendritic, circuitous, cul-de-sac-rich fabrics, a 2-km walk reaches as little as 32% of the people a 2-km circle would. That is still a finding about urban design. Where the network is poorly connected, the local environment of an average resident is smaller and therefore more homogeneous, which matters for everyday exposure to other groups (Netto et al., 2015). But it calls for a different framing, and for population-based (or population-matched) comparisons whenever network and Euclidean indices are contrasted.

Third, we find no support for configurational fragmentation, as we operationalise it, as a predictor of the gap. Neither the gap nor the share of reachable population is associated with F . What matters is the local texture of the network: dead ends, circuitry and meshedness. These are exactly the features that gated condominiums and closed *loteamentos* produce (Coy, 2006). One interpretation is that local-environment indices at a 2-km scale are sensitive to local permeability and blind to city-wide articulation, the property that synergy captures. Another is that our metric analogue does not capture what space-syntax synergy measures on axial maps. Testing CHASM-type segment-based measures (Saboya and Peres, 2025), which build the street grid into exposure itself, is a natural next step.

Limitations. First, OpenStreetMap coverage of informal settlements is uneven: stairways and alleys in favelas are often missing, which lengthens network paths and may inflate the gap there, so our estimates for hillside cities are best read as upper bounds. Second, tracts are represented

by a single interior point, and 1,020 of the 151,992 tracts lie more than 500 m from the nearest network node. Third, we consider the walking network only; public transport would lengthen reach and probably reduce the gap (Pereira et al., 2021; Bittencourt et al., 2021). Fourth, the units are municipalities, not functional urban areas, and results are subject to the modifiable areal unit problem. Finally, following the design of this paper, we do not test whether the gaps are larger than would be expected by chance. The random-labelling framework of Rey et al. (2021) and Knaap and Rey (2024) can be applied directly to our cached weights.

6 Conclusion

Measured with walking distance, racial segregation in the 91 largest Brazilian municipalities is uniformly higher than when measured as the crow flies, and rankings of the most segregated cities change. But the difference disappears once local environments are matched in population. The street network shapes measured segregation mainly by shrinking the neighbourhood rather than by separating groups within it. Dead ends and circuitry, not city-wide configurational fragmentation, are associated with that shrinkage. For measurement, this means that network and Euclidean indices should be compared at equal population scale. For planning, it points to the permeability of the local street fabric, the connectivity of peripheral *loteamentos* and gated enclaves, as the lever through which urban design conditions everyday exposure between racial groups. Future work should add inferential tests of the gap, multiscalar profiles, and public-transport networks.

Code and data availability

All code is in a self-contained companion repository (URL withheld during review) and reproduces every number, table and figure from public inputs. Census tract counts and geometries are from IBGE (2022 Census). Street networks are © OpenStreetMap contributors (ODbL); download dates are recorded per city. Unit tests verify, among other things, that our local-environment computation reproduces the PySAL `segregation w=` path to machine precision.

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A Robustness

Table 4: Network versus Euclidean indices under alternative specifications (91 municipalities). “Mean gap (%)” is the mean of $100(I^N - I^E)/I^E$; “Share gap > 0” is the share of cities in which the network index is higher. PP: *preta+parda*. Population-matched: Euclidean radii shrunk tract by tract to match the population of 2-km network environments.

Specification	Index	Mean gap	Mean gap (%)	Share gap > 0	Pearson	Spearman
2 km, linear	H (five groups)	0.005	29.835	1.000	0.980	0.984
2 km, linear	Dissimilarity (PP)	0.018	12.663	0.978	0.985	0.982
2 km, linear	Isolation (PP)	0.003	0.665	0.692	0.999	0.999
2 km, linear	Entropy (PP)	0.007	28.761	1.000	0.982	0.986
1 km, linear	H (five groups)	0.007	28.578	0.989	0.984	0.988
1 km, linear	Dissimilarity (PP)	0.017	10.965	0.978	0.989	0.984
1 km, linear	Isolation (PP)	0.004	1.152	0.802	0.999	0.999
1 km, linear	Entropy (PP)	0.008	24.604	0.978	0.987	0.988
3 km, linear	H (five groups)	0.005	36.901	1.000	0.983	0.985
3 km, linear	Dissimilarity (PP)	0.018	16.117	0.978	0.983	0.982
3 km, linear	Isolation (PP)	0.001	0.336	0.604	0.999	0.999
3 km, linear	Entropy (PP)	0.006	37.179	1.000	0.984	0.986
2 km, exponential	H (five groups)	0.005	28.481	0.989	0.983	0.986
2 km, exponential	Dissimilarity (PP)	0.017	11.861	0.967	0.987	0.984
2 km, exponential	Isolation (PP)	0.003	0.683	0.714	0.999	0.999
2 km, exponential	Entropy (PP)	0.006	26.950	0.989	0.985	0.988
Population-matched	H (five groups)	-0.000	-1.577	0.209	0.999	0.999
Population-matched	Dissimilarity (PP)	-0.001	-0.652	0.319	0.999	0.999
Population-matched	Isolation (PP)	-0.001	-0.265	0.286	1.000	1.000
Population-matched	Entropy (PP)	-0.000	-1.382	0.253	0.999	0.999

Table 5: Street-network topology across the 91 municipalities.

	count	mean	std	min	25%	50%	75%	max
Intersection density (per km2)	91.000	96.167	42.444	30.803	66.511	93.617	112.886	286.045
Street density (km/km2)	91.000	13.374	3.844	5.860	10.458	13.507	15.310	24.892
Mean street length (m)	91.000	85.027	12.286	50.293	76.284	84.081	93.440	110.996
Streets per node	91.000	2.878	0.164	2.334	2.773	2.880	2.995	3.254
Circuitry	91.000	1.045	0.016	1.018	1.033	1.043	1.054	1.112
Dead-end share	91.000	0.149	0.052	0.054	0.116	0.145	0.178	0.360
3-way share	91.000	0.636	0.055	0.468	0.593	0.645	0.679	0.732
4-way share	91.000	0.194	0.074	0.062	0.143	0.175	0.229	0.404
Cyclomatic number	91.000	9371.341	11411.163	1834.000	4142.000	6243.000	9468.000	86122.000
Meshedness	91.000	0.221	0.041	0.084	0.194	0.221	0.250	0.314
Gamma	91.000	0.480	0.027	0.389	0.462	0.480	0.499	0.542
F	91.000	0.552	0.152	0.283	0.420	0.542	0.680	0.896

Configurational fragmentation is estimated from random samples of 500 targets and 2,000 origins per city. Doubling both samples changes F by less than 0.02 in Porto Alegre, Curitiba and Salvador.

B Cross-check against the native PySAL network route

Using the `pandana` library, the PySAL `segregation` package can also build network-based local environments itself (Cortes et al., 2026). That route aggregates population at the network node

nearest to each tract and ignores the distance between the tract point and the node. For Porto Alegre, Curitiba and Salvador it yields network H within 0.001 of our estimates (Table 6).

Table 6: Network H from our tract-to-tract weights and from `segregation`’s native `pandana` route.

City	H, Euclidean	H, network (ours)	H, network (pandana)
Porto Alegre	0.0647	0.0745	0.0736
Curitiba	0.0410	0.0476	0.0472
Salvador	0.0310	0.0382	0.0382

C City-level results

City	H, Euclidean	H, network	Gap (%)	F
Florianópolis	0.024	0.049	108.987	0.627
Blumenau	0.009	0.019	98.939	0.370
Pelotas	0.016	0.027	73.877	0.668
Juiz de Fora	0.033	0.056	67.987	0.387
Carapicuíba	0.002	0.003	62.676	0.341
Ribeirão das Neves	0.001	0.002	52.246	0.535
Maceió	0.013	0.020	52.226	0.737
Diadema	0.005	0.008	52.102	0.541
Serra	0.008	0.012	51.846	0.794
Ananindeua	0.003	0.004	50.797	0.604
Ponta Grossa	0.014	0.022	50.716	0.365
Rio Branco	0.006	0.008	50.512	0.399
Itaquaquecetuba	0.002	0.003	48.926	0.643
São Gonçalo	0.006	0.008	48.858	0.561
Jundiaí	0.028	0.041	48.494	0.414
Caruaru	0.003	0.004	47.485	0.308
Jaboatão dos Guararapes	0.007	0.010	47.451	0.737
Taubaté	0.010	0.015	46.754	0.283
Paulista	0.003	0.004	45.872	0.644
Feira de Santana	0.006	0.009	44.053	0.352
Mauá	0.011	0.016	41.736	0.388
Vitória	0.055	0.078	41.150	0.722
Vitória da Conquista	0.009	0.012	39.770	0.391
São Bernardo do Campo	0.032	0.044	39.275	0.384
Canoas	0.012	0.016	39.116	0.504
Belford Roxo	0.002	0.003	39.074	0.837
Campina Grande	0.006	0.008	38.264	0.395
São José dos Pinhais	0.015	0.021	37.706	0.592
Praia Grande	0.019	0.026	36.760	0.704
Caxias do Sul	0.024	0.033	36.372	0.357
Macapá	0.004	0.005	35.155	0.598
Olinda	0.007	0.010	31.846	0.765
Cariacica	0.013	0.017	30.853	0.469
Rio de Janeiro	0.052	0.068	30.621	0.877
Petrolina	0.008	0.011	29.791	0.414
Recife	0.031	0.040	29.134	0.719
São Vicente	0.024	0.030	29.057	0.653
Duque de Caxias	0.005	0.006	28.477	0.896
Nova Iguaçu	0.011	0.014	28.329	0.692

Continued on next page

City	H, Euclidean	H, network	Gap (%)	F
Anápolis	0.013	0.016	27.118	0.454
Belém	0.012	0.015	26.333	0.585
Contagem	0.007	0.008	26.041	0.758
Suzano	0.026	0.033	25.976	0.494
Campos dos Goytacazes	0.042	0.052	25.748	0.483
Montes Claros	0.012	0.015	25.303	0.434
Barueri	0.028	0.035	25.209	0.564
Mogi das Cruzes	0.030	0.037	24.837	0.378
São Luís	0.017	0.022	24.293	0.542
Maringá	0.025	0.031	23.598	0.429
Belo Horizonte	0.055	0.068	23.561	0.594
Salvador	0.031	0.038	23.264	0.749
Franca	0.017	0.021	23.131	0.362
Betim	0.007	0.008	22.015	0.453
Vila Velha	0.031	0.038	21.852	0.580
Uberaba	0.023	0.028	21.662	0.449
Cuiabá	0.025	0.030	20.817	0.553
Aracaju	0.017	0.020	20.423	0.397
Sorocaba	0.023	0.027	20.049	0.337
Niterói	0.057	0.068	19.870	0.738
Fortaleza	0.017	0.021	19.495	0.563
João Pessoa	0.022	0.026	19.249	0.687
Cascavel	0.035	0.042	19.029	0.695
Uberlândia	0.028	0.034	18.436	0.439
Goiânia	0.035	0.041	17.824	0.574
São João de Meriti	0.004	0.005	17.682	0.785
Porto Velho	0.010	0.012	17.604	0.469
Londrina	0.054	0.063	16.990	0.520
Ribeirão Preto	0.046	0.054	16.330	0.380
Curitiba	0.041	0.048	16.257	0.435
Piracicaba	0.051	0.059	16.229	0.412
Joinville	0.022	0.026	16.098	0.575
Santo André	0.035	0.040	16.029	0.426
Aparecida de Goiânia	0.009	0.010	15.857	0.674
Bauru	0.036	0.042	15.675	0.502
Campinas	0.067	0.077	15.247	0.502
Porto Alegre	0.065	0.074	15.031	0.547
Boa Vista	0.010	0.011	14.911	0.636
Teresina	0.014	0.016	14.292	0.726
Osasco	0.018	0.021	13.971	0.430
São José dos Campos	0.040	0.045	13.827	0.509
Natal	0.015	0.016	13.090	0.729
Guarulhos	0.033	0.037	11.837	0.595
São Paulo	0.072	0.079	10.791	0.502
Campo Grande	0.035	0.039	10.096	0.569
Palmas	0.027	0.030	9.485	0.776
Caucaia	0.026	0.028	9.060	0.695
Santarém	0.011	0.012	7.579	0.335
São José do Rio Preto	0.034	0.036	5.132	0.394
Manaus	0.011	0.011	4.019	0.542
Brasília	0.043	0.044	2.716	0.844
Santos	0.041	0.041	0.751	0.809